

MacroPlate



Your AI-Powered Nutrition Coach & Personal Chef

Project Proposal

Track B Capstone - Mobile Application Development

IT 3rd Year | Information Technology

Hawassa University - Institute of Technology

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App Name : MacroPlate

Tagline: "Your AI-Powered Nutrition Coach & Personal Chef"

The problem?

In Ethiopia's rapidly growing fitness community, bodybuilders and gym enthusiasts face a critical gap between training and nutrition. While gyms and coaches are increasingly available across Addis Ababa and regional cities, there is no personalized nutrition guidance tailored to individual body metrics and fitness goals (cutting, maintaining, or bulking).

Current challenges include

Challenge	Description
Generic Advice	Most athletes rely on generic diet charts from the internet that don't account for individual body composition or activity levels
No Macro Tracking	Calculating daily protein, carb, and fat requirements requires nutrition knowledge that most fitness enthusiasts don't have
Time Constraints	Even when users know what to eat, meal preparation takes 2-3 hours daily - time that working professionals and students don't have
No-Delivery Services	There is no existing service in Ethiopia that bridges the gap between personalized nutrition planning and convenient food delivery

As a result, dedicated gym-goers spend months training without seeing results and it is not because they train poorly, but because they eat incorrectly. This leads to:

- Wasted gym membership fees
- Demotivation and dropout from fitness

- Potential health risks from improper nutrition

Brief User Persona

Primary Persona: Biruk

Attribute	Details
Name	Biruk
Age	26
Occupation	Software Engineer
Location	Addis Ababa
Fitness Goal	Bulk (Build muscle)
Pain Points	Confused about protein requirements, no time for meal prep, wants science-based nutrition

Our primary user is Biruk, a university student and gym enthusiast in Addis Ababa. And he says, *“I train 5 days a week but I'm not seeing the muscle growth I expected. I know nutrition is important, but I don't have time to research and prep meals. I need someone to tell me exactly what to eat and bring it to me.”*

Secondary Persona: Meron

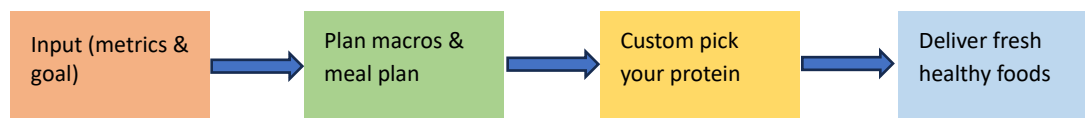
Attribute	Details
Name	Meron
Age	24
Occupation	Fitness Influencer & Women's Wellness Coach
Location	Addis Ababa
Fitness Goal	Cut (Weight loss for clients)
Pain Points	Cannot provide practical meal plans to clients, needs tool to scale her coaching

Our other user is Meron, struggles to help her clients and she also says, *“My female clients want to lose weight but don't know how to eat. I give them advice, but they struggle to implement it. I need a tool that creates simple, personalized meal plans I can share with them.”*

The Solution?

MacroPlate is a mobile application that acts as a personalized nutritionist and meal prep service. The user begins by selecting their goal (Cut / Maintain / Bulk) and providing their body metrics (age, weight, height, activity level). Our macroplate IA (intelligent algorithm) calculates daily calorie and macro targets (protein, carbs, fats) and generates a custom weekly meal plan that fits those numbers.

How it works?



Key Benefits:

Benefit	Description
Personalized	Meals based on YOUR body metrics (age, weight, height, activity)
Science-Based	Uses BMR formula and proven macro splits
Flexible	Swap proteins while maintaining macro targets
Convenient	No cooking, no shopping, no planning
Affordable	Bulk meal prep pricing, delivered weekly

Core features (MVP)

Feature 1: User Authentication & Onboarding

Sub-feature	Description
Sign Up	Email/password registration with OTP verification
Login	Email/password and Google sign-in options
Password Reset	Forgot password with email reset link
Onboarding Flow	Welcome screen, goal selection (Cut/Maintain/Bulk), permission requests

Why this is essential: Users need secure accounts to save their meal plans and orders.

Feature 2: User Profile & Body Metrics Input

Sub-feature	Description
Body Metrics	Age, weight (kg), height (cm), gender
Activity Level	5 options: Sedentary, Light, Moderate, Very Active, Athlete
Goal Confirmation	Cut (deficit), Maintain (maintenance), Bulk (surplus)

Why this is essential: Accurate macro calculation requires accurate input data.

Feature 3: Macro Calculation & Meal Plan Generation

Sub-feature	Description
BMR Calculation	Harris-Benedict formula for basal metabolic rate
TDEE Calculation	$BMR \times \text{Activity Factor}$
Goal Adjustment	Cut: -500 cal, Maintain: 0 cal, Bulk: +300 cal
Macro Split	Protein: 2.0-2.2g per kg, Fats: 25% of calories, Carbs: remainder
7-Day Meal Plan	Breakfast, Lunch, Dinner, Snack for each day

Why this is essential: This is the core intelligence of MacroPlate.

Feature 4: Meal Customization ("Pick Your Protein")

Sub-feature	Description
Customize Button	Available on every meal card
Alternative Proteins	Chicken, beef, fish, tofu, eggs with macros
Macro Recalculation	Daily totals update automatically after swap
Price Difference	Shows extra cost or savings per swap

Why this is essential: This is MacroPlate's key differentiator - flexibility within structure.

Feature 5: Order & Delivery Scheduling

Sub-feature	Description
Order Summary	Meal quantities, subtotal, delivery fee, total
Delivery Date	Calendar picker (minimum tomorrow, max 14 days)
Time Slot	Morning, Afternoon, Evening options
Delivery Address	Saved addresses + add new address
Payment Method	Cash on Delivery, Telebirr, Chapa
Order Confirmation	Order number, delivery details, total

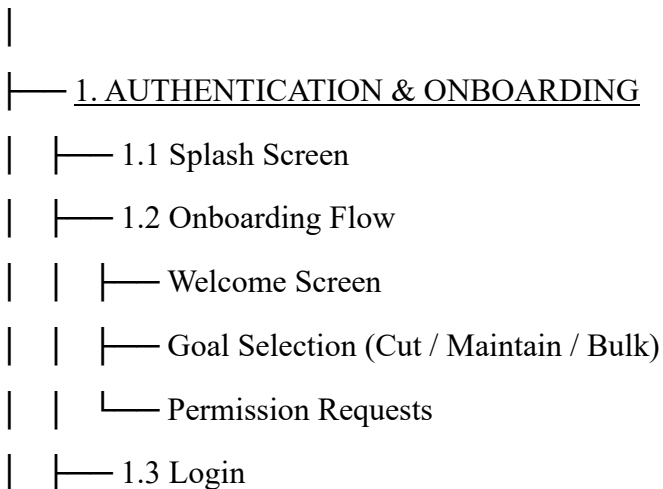
Why this is essential: This completes the loop from planning to delivery.

Future Features (Roadmap)

Phase	Feature	Description
Phase 2	Real Payment Integration	Telebirr and Chapa API integration for in-app payments
Phase 2	Meal Logging	Users log what they ate, see remaining macros
Phase 2	Progress Tracking	Weight, measurements, progress photos over time
Phase 3	Fitness App Integration	Connect to Google Fit, Fitbit to auto-adjust calories
Phase 3	Barcode Scanner	Scan grocery items to see if they fit macro targets
Phase 3	Social Features	Share meal plans, compete with friends
Phase 4	AI Food Recognition	Take photo of meal, app estimates macros
Phase 4	Restaurant Partnerships	"MacroPlate Approved" meals at local restaurants

Information Architecture

MACROPLATE - INFORMATION ARCHITECTURE



- | | |— Login with Email
- | | |— Login with Google
- | | |— Forgot Password
- | | |— Sign Up Link
- | |— 1.4 Sign Up
- | |— Sign Up Form
- | |— Email Verification (OTP)
- | |— Profile Setup Redirect

|— 2. USER PROFILE INPUT

- | |— 2.1 Body Metrics Form
- | | |— Age, Weight, Height, Gender
- | |— 2.2 Activity Level Selection
- | | |— Sedentary, Light, Moderate, Very Active, Athlete
- | |— 2.3 Goal Confirmation
- | | |— Cut, Maintain, Bulk
- | |— 2.4 Submit Button

|— 3. MAIN DASHBOARD

- | |— 3.1 Header (Greeting, Notifications, Profile)
- | |— 3.2 Daily Summary Card (Calories, Protein, Carbs, Fats)
- | |— 3.3 Quick Actions (View Plan, Order This Week)
- | |— 3.4 Today's Meals (Breakfast, Lunch, Dinner, Snack)

|— 4. MEAL PLAN MODULE

- | |— 4.1 Weekly View (Day Tabs: Mon - Sun)
- | |— 4.2 Day Cards (Breakfast, Lunch, Dinner, Snack)

- | └─ 4.3 Meal Detail View (Ingredients, Macros)

- |

- | └─ 5. CUSTOMIZATION MODULE ("Pick Your Protein")

- | | └─ 5.1 Current Meal Display

- | | └─ 5.2 Alternative Proteins List (with macros & price)

- | | └─ 5.3 Confirm Changes Button

- | | └─ 5.4 Cancel Button

- |

- | └─ 6. ORDER & DELIVERY

- | | └─ 6.1 Order Summary (Quantities, Total)

- | | └─ 6.2 Delivery Scheduling (Date, Time Slot)

- | | └─ 6.3 Delivery Address (Saved, Add New)

- | | └─ 6.4 Payment Method (Cash, Telebirr, Chapa)

- | | └─ 6.5 Place Order Button

- | | └─ 6.6 Order Confirmation

- |

- | └─ 7. USER PROFILE

- | | └─ 7.1 Profile Header (Photo, Name, Email)

- | | └─ 7.2 Body Metrics Section (Edit Metrics)

- | | └─ 7.3 Plan History

- | | └─ 7.4 Saved Preferences

- | | └─ 7.5 Account Settings

- |

- | └─ 8. NOTIFICATIONS

- | | └─ Order Confirmation, Delivery Reminder

- | | └─ Weekly Plan Ready, Payment Reminder

- | | └─ Meal Prep Reminder

|

└─ 9. ERROR STATES

└─ Network Error (No Internet, Retry)

└─ Form Validation Errors

└─ Authentication Errors

└─ Order Errors

Technical Stack

Layer	Technology	Purpose
Frontend	Android (Kotlin/Java)	Mobile app development
UI Framework	Jetpack Compose or XML	User interface
Backend	Firebase / Node.js	User authentication, data storage
Database	Firebase Firestore / Room	User data, meal plans, orders
Authentication	Firebase Auth	Email/Google login
Payments	Telebirr API, Chapa API	Payment processing (Phase 2)
Version Control	Git & GitHub	Code management
Project Management	Jira	Agile task tracking
Design	Figma, FigJam	Wireframes, IA, workflows

Project Resources

Resource	Link
GitHub Repository	MAD-Project-MacroPlate
Figma Wireframes	View Wireframes

Resource	Link
FigJam IA & Workflows	View FigJam
Jira Board	View Jira

Team Members & Roles

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|-------------------------------|----------------------------------|
| 1. UI/UX Implementation Lead | Helina Betre |
| 2. Database & State Lead | Yonas Atrsie |
| 3. Core Logic & Services Lead | Nebiyu Daniel |
| 4. Network & API Lead | Miki Girma |
| 5. DevOps & Testing Lead | Yishak Tadele & Bilisuma Tsegaye |